

Responsible and Trustworthy AI

Course Metadata

Credit Structure	L-T-P: 4-0-0 Total: 4
Offering Semester	Spring, 6 th semester
Level of Subject	UG Level
Compulsory or elective?	Core
Pre-requisite	Deep Learning Foundations, Introduction to Machine Learning
Faculty Names	Koustav Rudra, Jiaul H Paik
Overlapping Subjects	AI and Ethics (CS60016) ~20% Dependable and Secure AI-ML (AI60006) ~5%
Justification of overlap	This course will be a core subject for BTech program in AI

Course Description

Web content analysis using text has been recently dominated by complex representation learning approaches using neural models. A key concern using complex learning systems is regarding their interpretability in that it is hard to determine if the predictions of these models are grounded in the right reasons. Towards this, there has been an upsurge of approaches that intend to incorporate trustworthiness in the decisions of complex machine and deep learning models. These trustworthiness factors cover several aspects such as interpretability, fairness, privacy, robustness, and security. Interpretability tries to provide an explanation for model predictions whereas fairness tries to reduce bias in the model prediction. Privacy provides new learning techniques so that model should not reveal the particulars of training data to users. Finally, robustness and safety take care of deep learning models resistance against attacks.

Considering the rapid adoption of deep learning and its effectiveness to different areas such as text, image, video, speech, etc., this course is proposed to develop the theoretical foundation of trustworthiness aspects of machine learning models and their use in real world applications in diverse domains.

Course Content

- Introduction to responsible AI
- Model agnostic interpretable methods
- Neural network interpretation
- Algorithmic bias and Risk assessment
- Privacy aware learning and Model robustness
- Model safety and accountability

- Application of Responsible AI

Lecture-wise Breakup

Sl No.	Module	Unit	Module Hr (48)
1	Introduction	Basic of deep learning models	2
		Requirement of responsible AI	
		Types of interpretability, Evaluation techniques	
2	Model Agnostic Interpretability Methods	Global model agnostic methods: Partial dependence plot, Feature interaction	6
		Local model agnostic methods: ICE, LIME, SHAP	
3	Neural Network Interpretation	Pixel attribution methods: Saliency maps, Grad-CAM	6
		Influential instances, Concept detection	
4	Algorithmic Bias	Types of bias in machine learning algorithm	6
		Mitigating bias in learning algorithm, Evaluation of bias	
5	Risk Assessment	Analysis of COMPAS recidivism algorithm	5
		ML algorithm for bail and sentencing	
6	Privacy Aware Learning	Privacy in ML models	7
		Data augmentation based privacy	
		Federated learning	
7	Model Robustness	Adversarial attack	7
		Robust model design via loss function modification, adversarial training	
		Influence Functions, data points	
8	Model Safety and Accountability	Social and Fiscal impact assessment	4
		System failure and accountability	
9	Application of Responsible AI on NLP and vision domain	Interpretable methods for entity detection, summarization, image classification	5
		Fair model for credit score prediction	
		Robust modelling in cancer prediction	

Books

1. **Interpretable Machine Learning: A Guide for Making Black Box Models Explainable**, Christoph Molnar, ISBN: 9780244768522, 2020 (2nd edition)
2. **AI and Ethics: A Computational Perspective**, Animesh Mukherjee, IOP Publishing Ltd, ISBN: 978-0-7503-6116-3, 2023
3. **Deep Learning**, Ian Goodfellow and Yoshua Bengio and Aaron Courville, MIT Press, 2016

References

1. V. Sehwal, AN. Bhagoji, L. Song, C. Sitawarin, D. Cullina, M. Chiang, P. Mittal. "Analyzing the robustness of open-world machine learning". In Proceedings of the 12th ACM Workshop on Artificial Intelligence and Security 2019 Nov 11 (pp. 105-116).
2. JG. Lee, Y. Roh, H. Song, SE. Whang. "Machine learning robustness, fairness, and their convergence". In Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining 2021 Aug 14 (pp. 4046-4047).
3. N. Mehrabi, F. Morstatter, N. Saxena, K. Lerman, A. Galstyan. "A survey on bias and fairness in machine learning". ACM computing surveys (CSUR). 2021 Jul 13;54(6):1-35.
4. A. Madsen, S. Reddy, S. Chandar. "Post-hoc interpretability for neural nlp: A survey. ACM Computing Surveys". 2022 Dec 23;55(8):1-42.
5. S. Luo, H. Ivison, C. Han, J. Poon. "Local interpretations for explainable natural language processing: A survey". arXiv preprint arXiv:2103.11072. 2021 Mar 20.